

it was iterated that, although FPGA vendors would continue to emphasise efficient IP cores for 40- and 100-Gbit Ethernet, Fibre Channel Gen 5 and PCI Express Gen 3, they will definitely be working with EDA specialists such as Cadence, Synopsys and Mentor to offer power-optimised on-chip buses, both for memory interconnect and for links between multi-processor cores.

Embedded networks-on-chips for FPGAs. In February this year, Mohamed Abdelfattah and others at the University of Toronto proposed that, integrating networks-on-chips (NoCs) on FPGAs can improve device scalability and facilitate design by abstracting communication and simplifying timing closure, not only between modules in the FPGA fabric but also with large hard blocks, such as high-speed I/O interfaces. Their design features mixed and hard NoCs, which add less than one per cent to the area of the FPGAs yet run five to six times faster than the soft equivalents.

Their research showed that hard NoCs consume 4.5-10.4mJ of energy per gigabyte of data transferred, which is comparable to the energy efficiency of simple soft point-to-point links in FPGAs, while offering greater benefits. The design is more energy-efficient and space-saving compared to custom buses configured into the fabric by commercial system integration tools.

3PLDs from Tabula. FPGA player Tabula has come up with future-ready spacetime technology, which apparently overcomes the limitations of FPGA's 2D topologies by using a third dimension to create a class of 3D programmable logic devices (3PLDs). Spacetime uses hardware capable of dynamically reconfiguring at multi-gigahertz rates and the Stylus compiler, which trans-

Some state-of-the-art FPGAs...

- Xilinx's Virtex range of 20nm, TSMC-fabbed, interposer-based, 4.4-million-logic-cell UltraScale devices with seamless footprint migration to 16nm FinFET
- Lattice Semiconductor's tiny 1.4mm×1.4mm BGA, 25-microwatt, 384-cell iCE40 device, which sells at a bulk price less than 50 cents!
- Altera's generation 10 FPGAs, including the TSMC 20nm Arria 10 and the 3D-capable, Intel 14nm Tri-Gate Stratix 10
- QuickLogic's customer-specific standard product (CSSP) devices
- Microsemi's SmartFusion2 SoC FPGAs and IGL002 FPGAs
- Achronix's 22nm, FinFET-based, million-LUT, HD1000 Speedster
- Tabula's Spacetime devices

parently manages this ultra-rapid reconfiguration of logic, memory and interconnect resources throughout the chip. Actually, the third dimension of space is created using time, and hence the name Spacetime!

According to Tabula, "A Spacetime device reconfigures on the fly at multi-gigahertz rates, executing each portion of a design in an automatically defined sequence of steps. Manufactured using a standard CMOS process, Spacetime uses this ultra-rapid reconfiguration to make time a third dimension. This results in a 3D device with multiple layers or folds in which computation and signal transmission can occur. Each fold performs a portion of the desired function and stores the result in place. When some or all of a fold is reconfigured, it uses the locally stored data to perform the next portion of the function.

By rapidly reconfiguring to execute different portions of each function, a 3D Spacetime device can implement a complex design using only a small fraction of the resources that would be required by an FPGA, with its inherently 2D architecture." The big plus is the manageable price-point of Spacetime devices.

Beyond programmability, self-learning processors. Programmable hardware is the future but cognitive

computing is even more exciting. You will be surprised to know that, sometime this year, computers that learn from their own mistakes will be commercially available!

Stanford University, IBM, Google and several others are working on this. Such self-learning computers will enable extreme automation, fault tolerance and adaptability, making computer crashes a thing of the past. They will use a new class of processors, which comprise electronic components that can be connected by wires that mimic biological synapses. Because of the neuron-like elements in these processors, they are known as 'neuromorphic processors.' These processors need not be programmed; instead the connections in the circuits are formed based on correlations in data that the processor has learned earlier. As data flows, the connections might get further altered, leading to changes in behaviour just as in the case of humans.

According to a comment by IBM scientist Dharmendra Modha, this methodology brings computation to data instead of the other way round. "Sensors become the computer, and it opens up a new way to use computer chips that can be everywhere," he said in an earlier report. When they first arrive, these computers will work as co-processors in devices such as smartphones rather than as standalone processors. Qualcomm, which has developed such a processor together with IBM, plans to release it commercially sometime this year. Stanford's Brains in Silicon project is another major contender in this space. ●

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